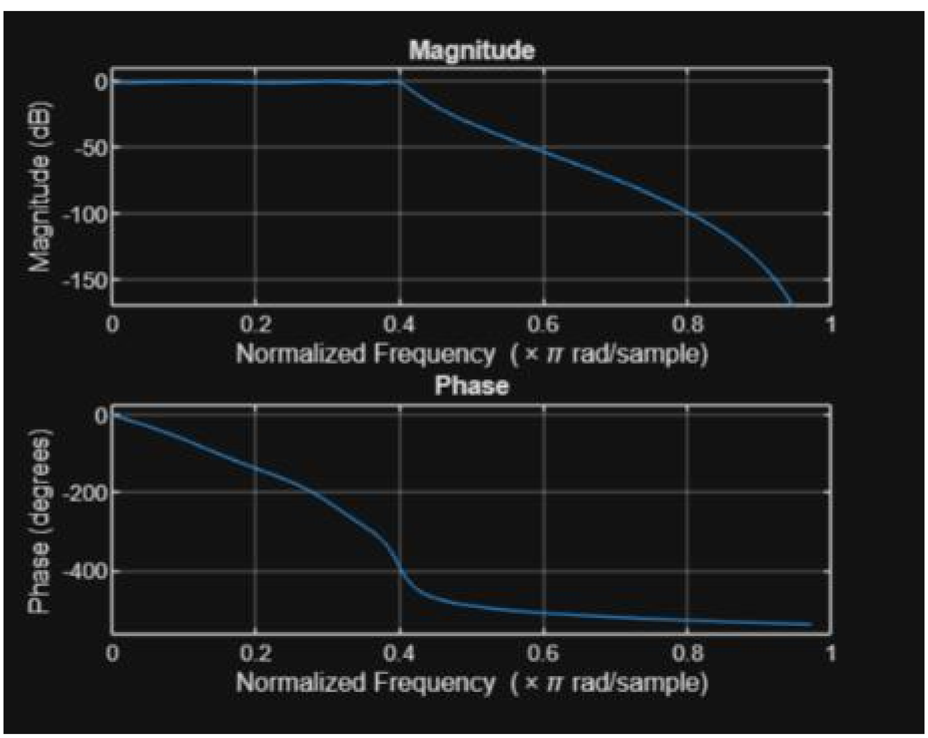
EX NO:2 Chebyshev IIR Filter

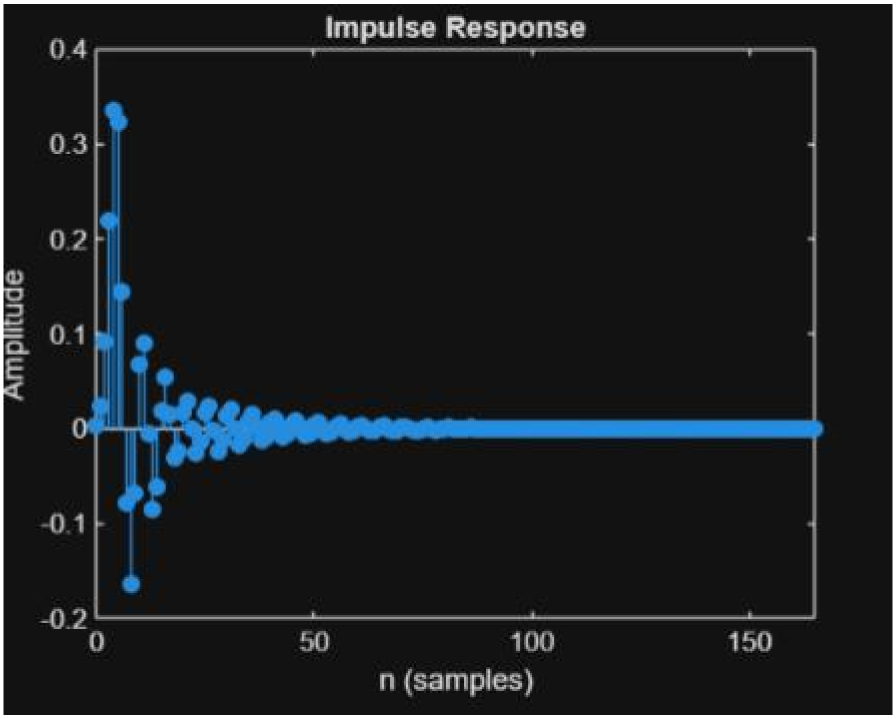
2.1. Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass Filter using MATLAB

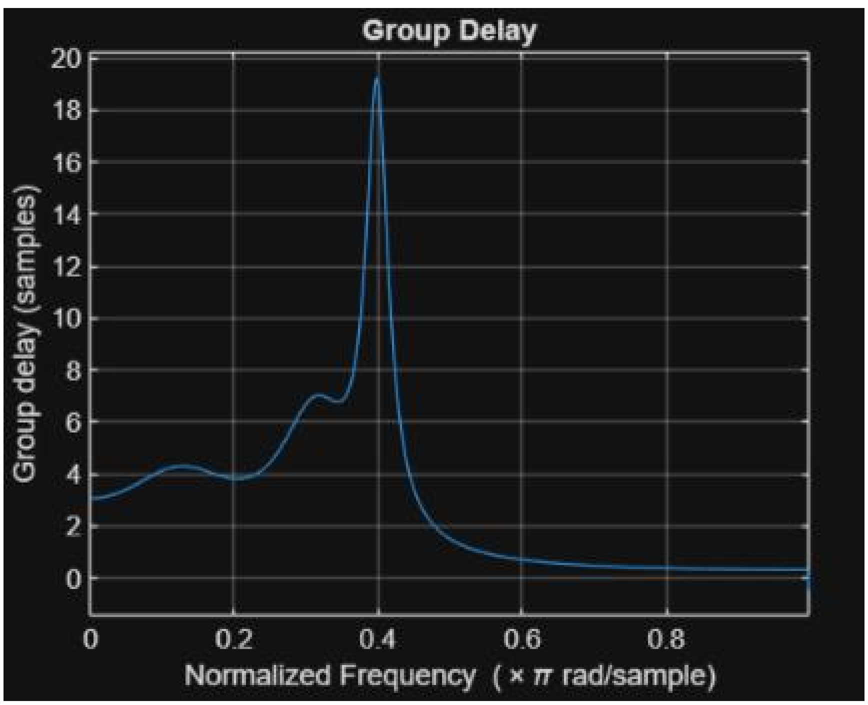
dsp exp 1.2.mlx | dsp exp 2.1.mlx | dsp exp1.1.mlx

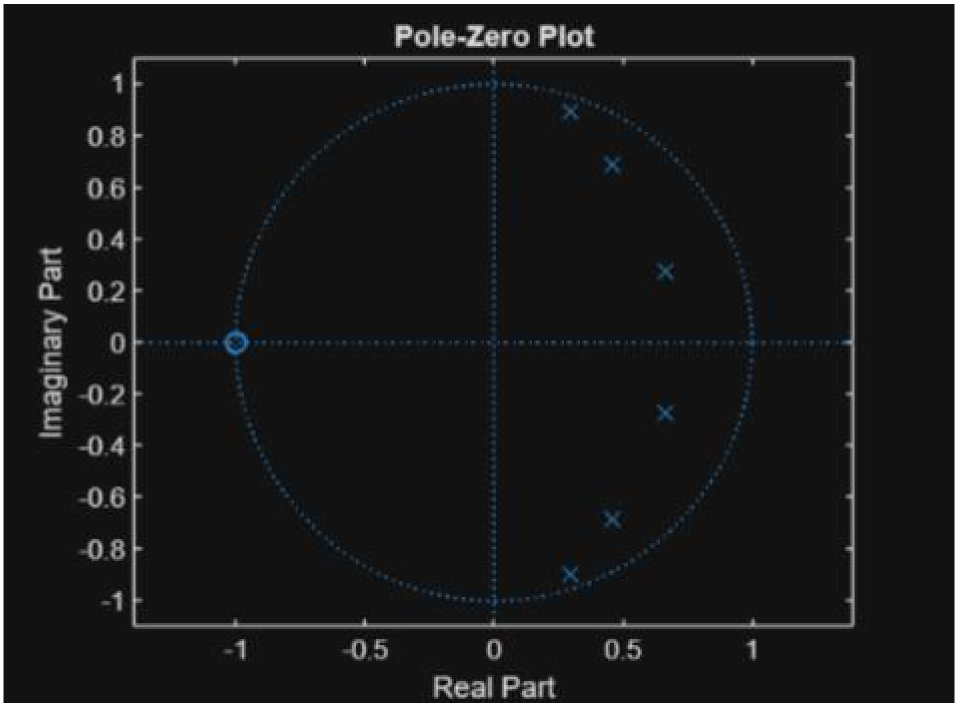
/MATLAB Drive/dsp exp 2.1.mlx

```
clc; clear;

Wp=0.4; Ws=0.55; Rp=1; As=40;

[N,Wn]=cheb1ord(Wp,Ws,Rp,As);
[b,a]=cheby1(N,Rp,Wn);

fprintf('Filter Order = %d\n',N);
fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);

if all(abs(roots(a))<1)
    disp('System is STABLE');
else
    disp('System is UNSTABLE');
end

% -------- GRAPHS --------
figure; freqz(b,a); % Magnitude + Phase
figure; impz(b,a); % Impulse Response
figure; grpdelay(b,a); % Group Delay
figure; zplane(b,a); % Pole-Zero
```




Impulse Response
Amplitude
n (samples)
0.4
0.3
0.2
0.1
0
-0.1
-0.2
0
50
100
150


Group Delay
Group delay (samples)
Normalized Frequency (× π rad/sample)
20
18
16
14
12
10
8
6
4
2
0
0
0.2
0.4
0.6
0.8



Result:

1. The Chebyshev Type-I IIR High-pass filter was successfully designed, and its magnitude response, phase response, impulse response, and group delay were analyzed.

2. The impulse response and group delay characteristics were obtained and found to satisfy the given specifications.

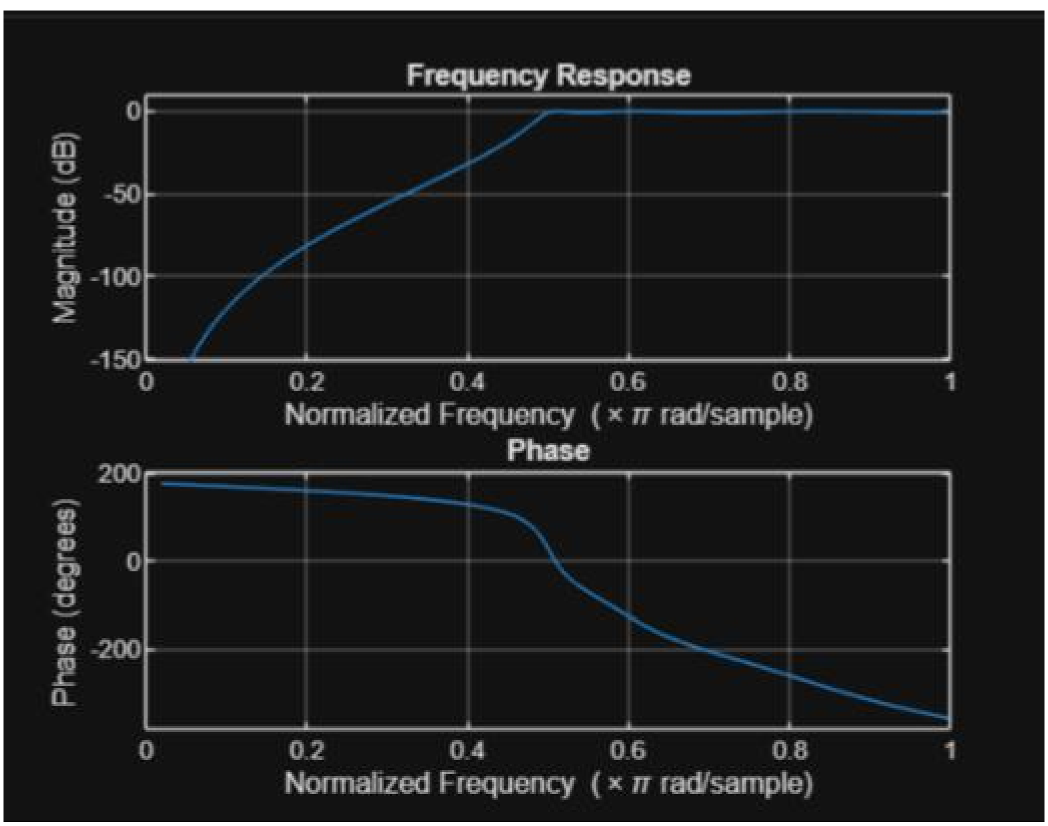
3. The pole-zero plot confirmed that all poles lie inside the unit circle, verifying the stability and proper performance of the filter.

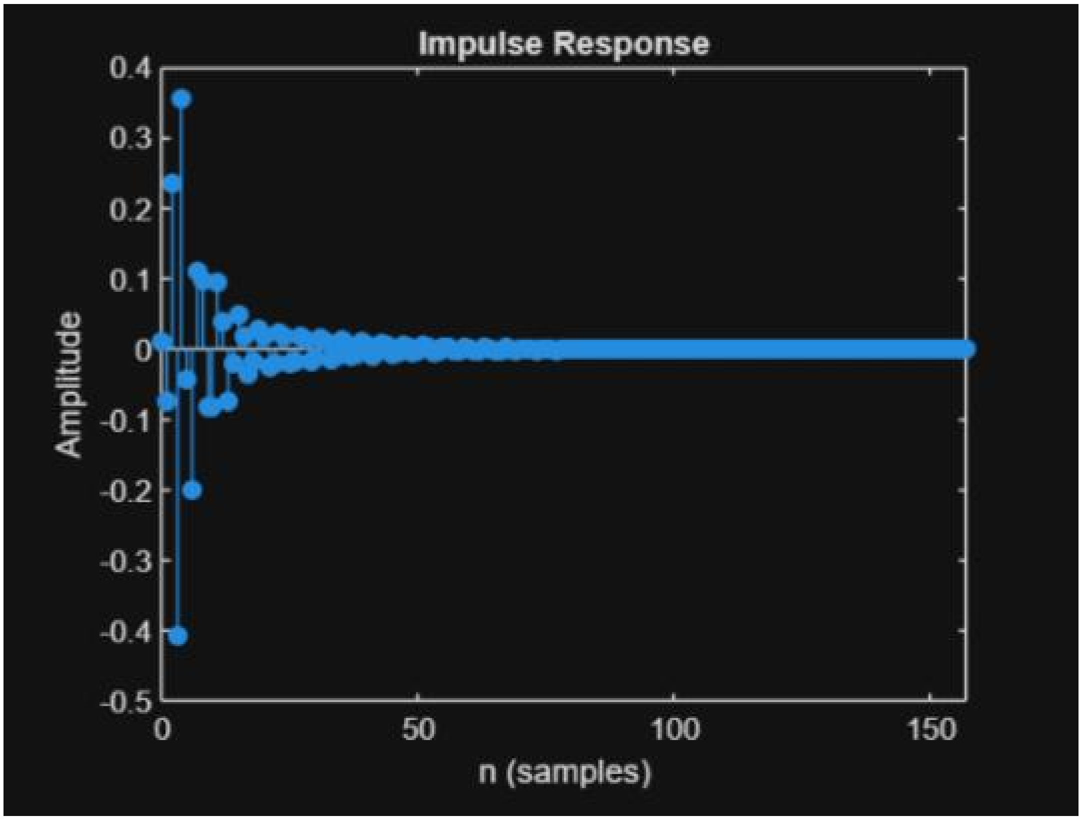
4. The minimum filter order and cutoff frequency were successfully determined and found to satisfy the given design specifications.

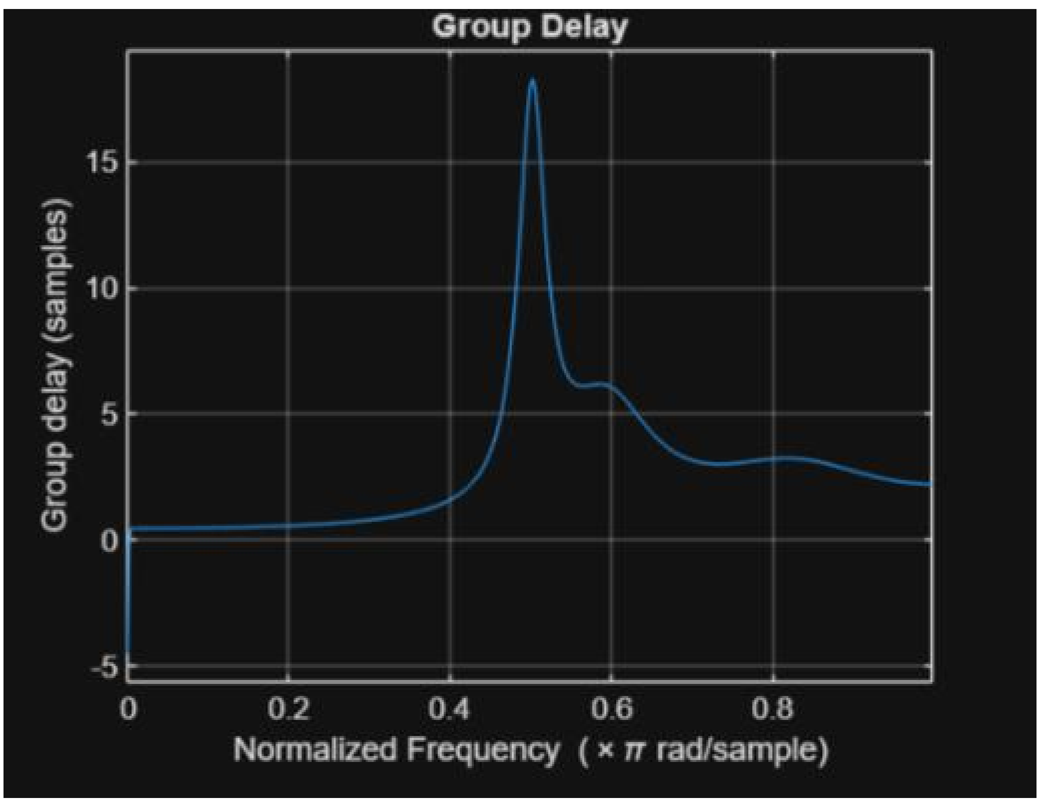
2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass Filter

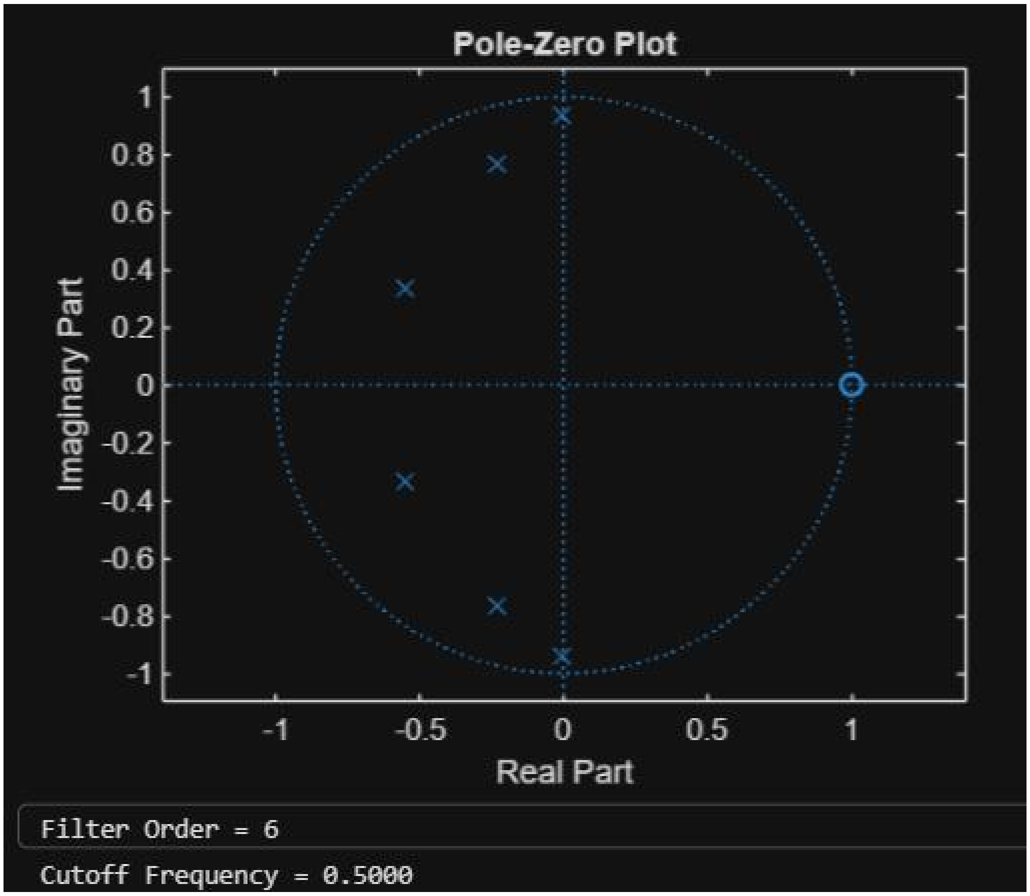
/MATLAB Drive/dsp exp 2.2.mlx

```
clc;
clear;
close all;
% Specifications
Rp = 1; % Passband Ripple (dB)
Rs = 40; % Stopband Attenuation (dB)
Wp = 0.5; % Passband Edge
Ws = 0.35; % Stopband Edge
% Find Filter Order
[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);
% Design High-Pass Filter
[b,a] = cheby1(n,Rp,Wn,'high');
% Frequency Response
figure;
freqz(b,a);
title('Frequency Response');
% Impulse Response
figure;
impz(b,a);
title('Impulse Response');
% Group Delay
figure;
grpdelay(b,a);
title('Group Delay');
% Pole-Zero Plot
figure;
zplane(b,a);
title('Pole-Zero Plot');
fprintf('Filter Order = %d\n',n);
fprintf('Cutoff Frequency = %.4f\n',Wn);
```




Impulse Response
Amplitude
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
-0.5
0
50
100
150
n (samples)


Group Delay
Group delay (samples)
15
10
5
0
-5
0
0.2
0.4
0.6
0.8
Normalized Frequency (× π rad/sample)



Result:

1. The Chebyshev Type-I IIR low-pass filter was successfully designed, and its magnitude response, phase response, impulse response, and group delay were analyzed.

2. The impulse response and group delay characteristics were obtained and found to satisfy the given specifications.

3. The pole-zero plot confirmed that all poles lie inside the unit circle, verifying the stability and proper performance of the filter.

4. The minimum filter order and cutoff frequency were successfully determined and found to satisfy the given design specifications.